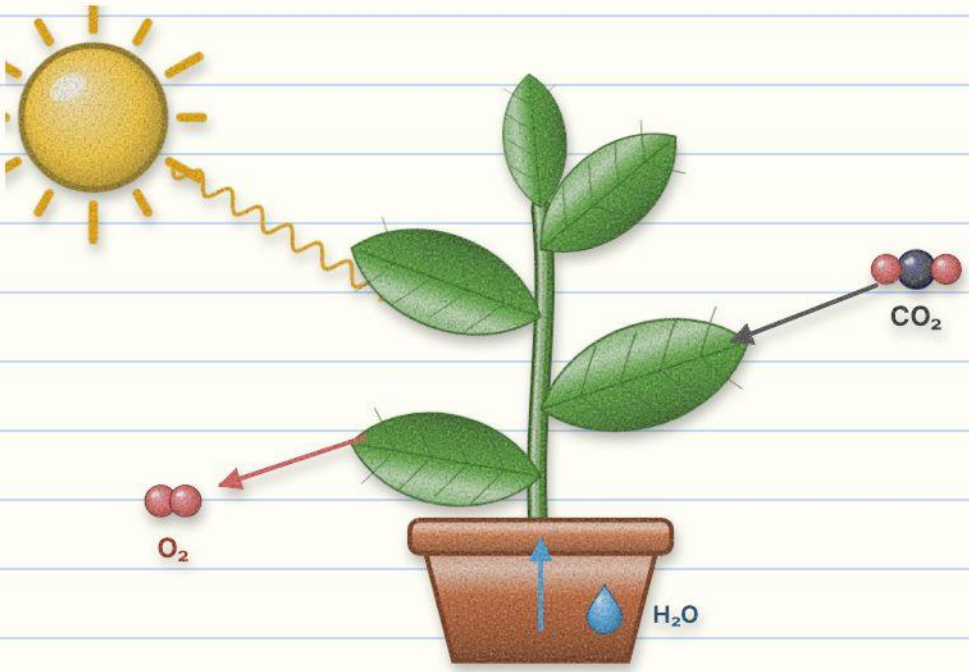
Photosynthesis in Higher Plants

PU-I Biology · Chapter 11

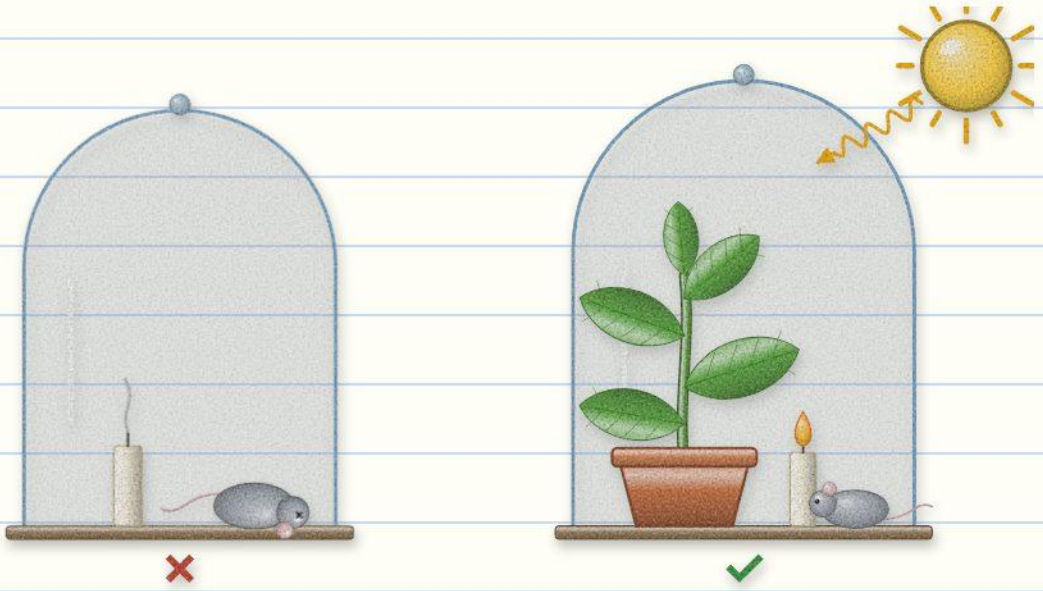
Handwritten notes by Narendra Sir

Lecturer in Biology · Founder/CEO, BAIO Learning System



I. Early Experiments

- ★ Photosynthesis = physico-chemical process; light energy → organic compounds. Source of all food & O_2.
- Priestley (1770): mint plant restores air fouled by candle / mouse.
- Ingenhousz: needs sunlight; only green parts give O_2.
- Sachs (1854): glucose → starch; chlorophyll in chloroplasts.
- ★ Engelmann: *Cladophora* + aerobic bacteria → first action spectrum (blue & red).

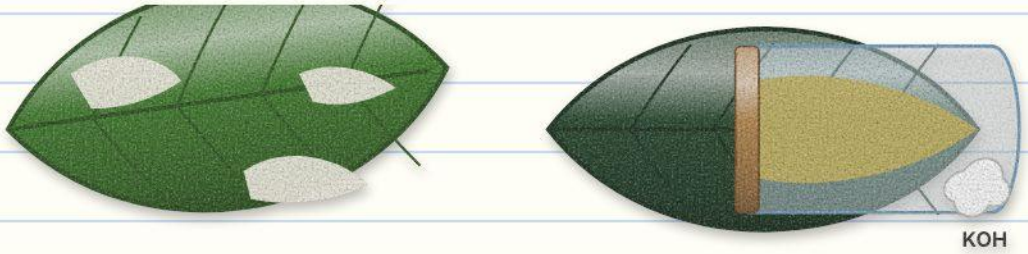


Ib. Oxygen comes from Water

- ★ van Niel (purple & green bacteria): $2H_2A + CO_2 \rightarrow 2A + CH_2O + H_2O$
- Sulphur bacteria use H_2S → give sulphur, not O_2 ⇒ in plants O_2 comes from H_2O.
- Proved with ^{18}O (Ruben & Kamen).

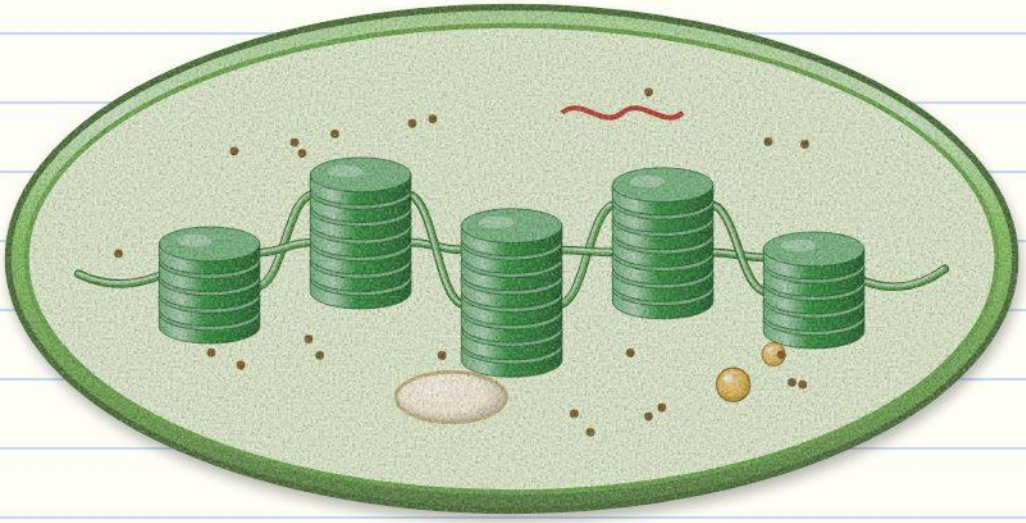
★ $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2$

- Half-leaf expt: KOH absorbs CO_2 → no starch there.



Starch ✓ / No starch

2. Chloroplast & Pigments



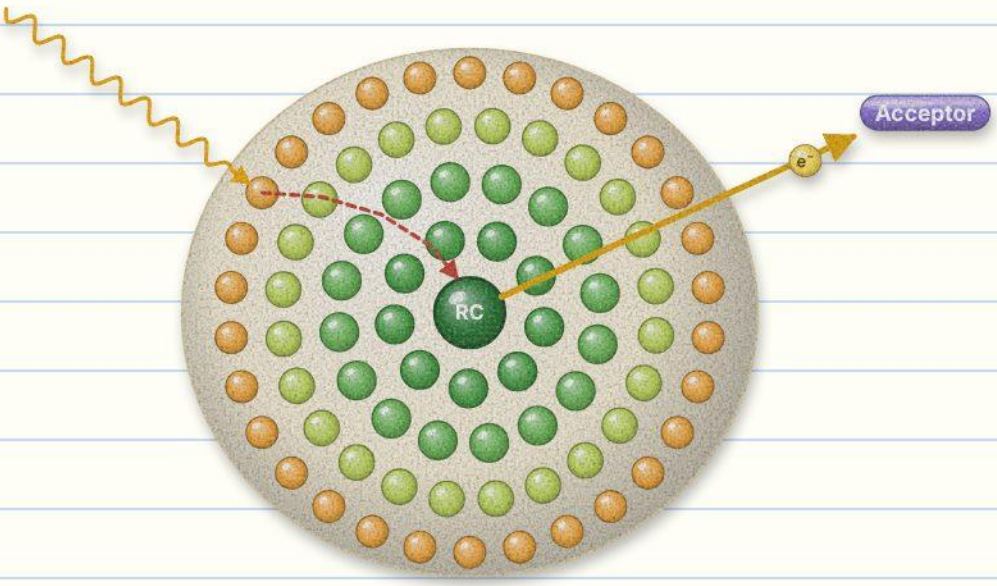
- ★ Grana / thylakoids → light reaction → ATP + NADPH.
- Stroma → enzymes fix CO_2 → sugar ("dark" reaction — not directly light driven).

Pigment	Colour
Chl a	blue-green (chief)
Chl b	yellow-green
Xanthophyll	yellow
Carotenoid	yellow-orange

★ Max absorption & photosynthesis in blue & red.

3. Light Reaction

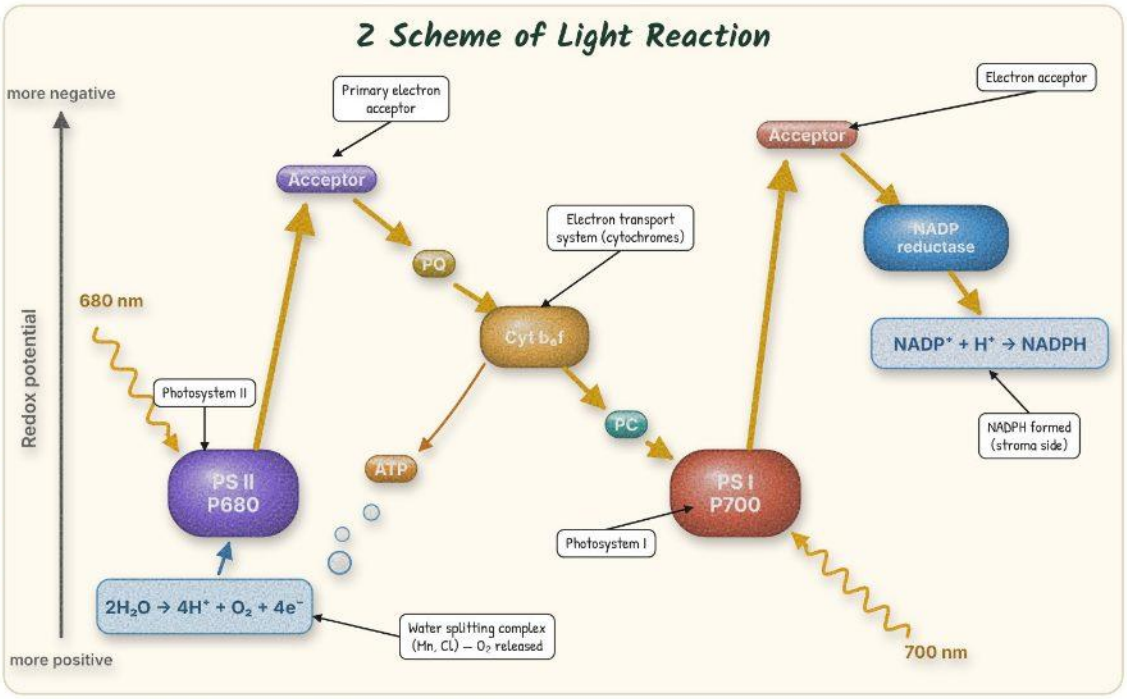
- Light absorption, water splitting, O_2 release, ATP + NADPH.
- ★ LHC = antenna pigments + one chl a reaction centre.
- ★ PS I → P700 · PS II → P680 (named in order of discovery).



★ Photolysis: $2H_2O \rightarrow 4H^+ + O_2 + 4e^-$ — with PS II, needs Mn, Cl.

Hill reaction (1939): isolated chloroplasts + acceptor → O_2 without CO_2.

4. Z Scheme



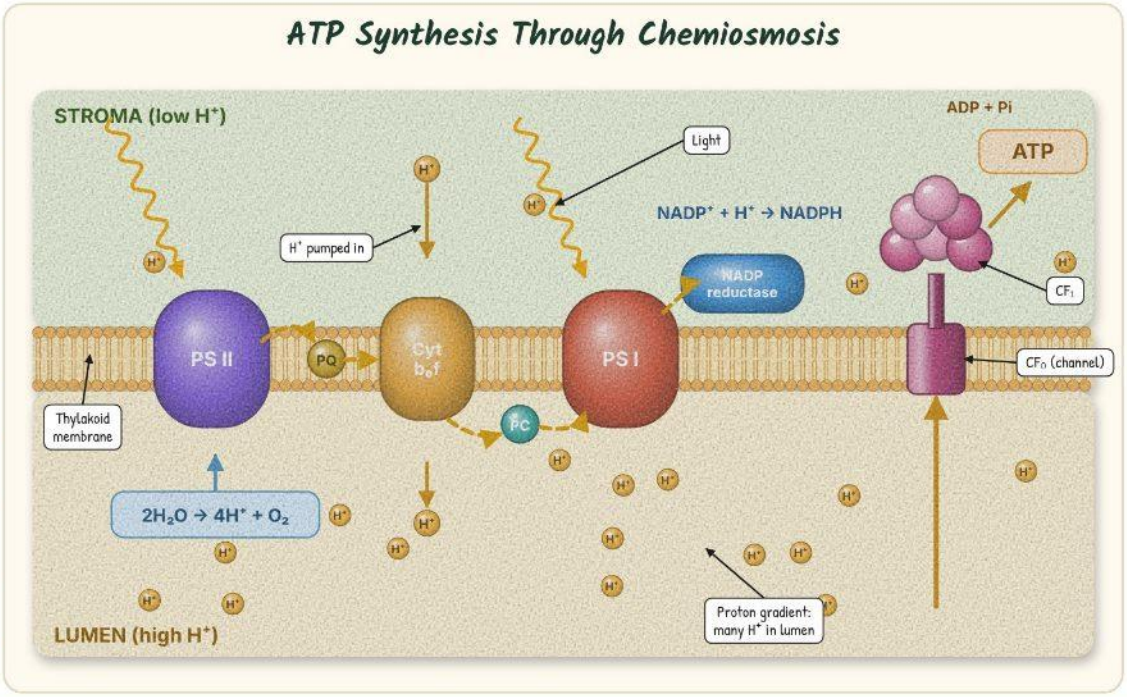
- ★ H_2O → P680 → acceptor → ETS → P700 → acceptor → NADP reductase → NADPH
- Gives ATP + NADPH + O_2 = non-cyclic photophosphorylation.

Cyclic

- ★ PS I only, stroma lamellae, light > 680 nm → only ATP (no NADPH, no O_2).

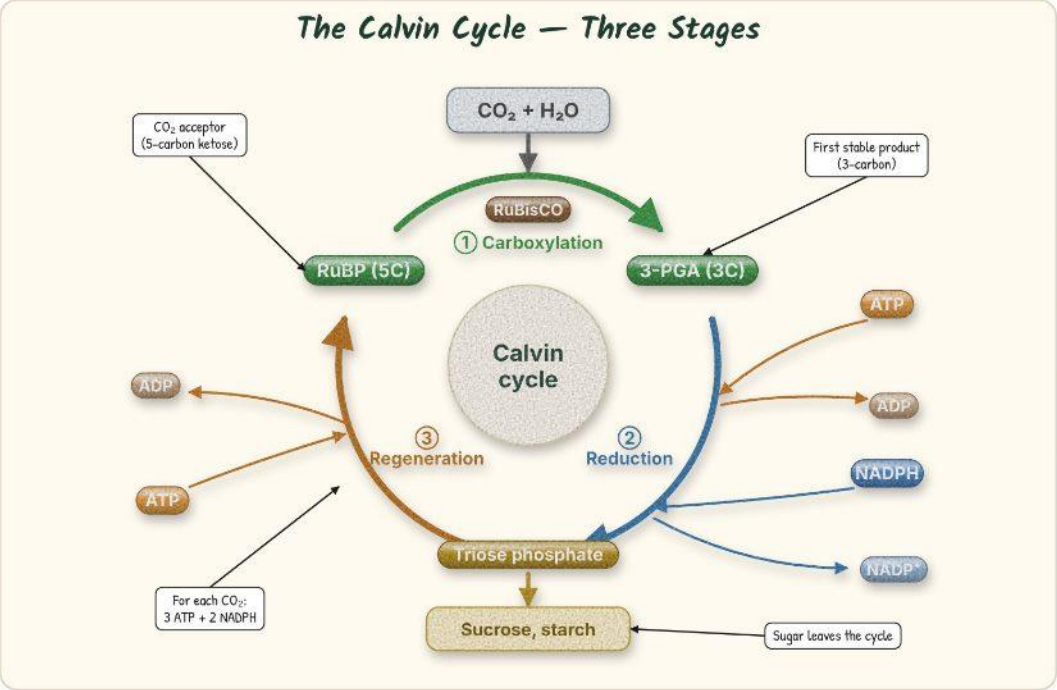
5. Chemiosmosis

★ Peter Mitchell (1961): ATP made by a proton gradient.



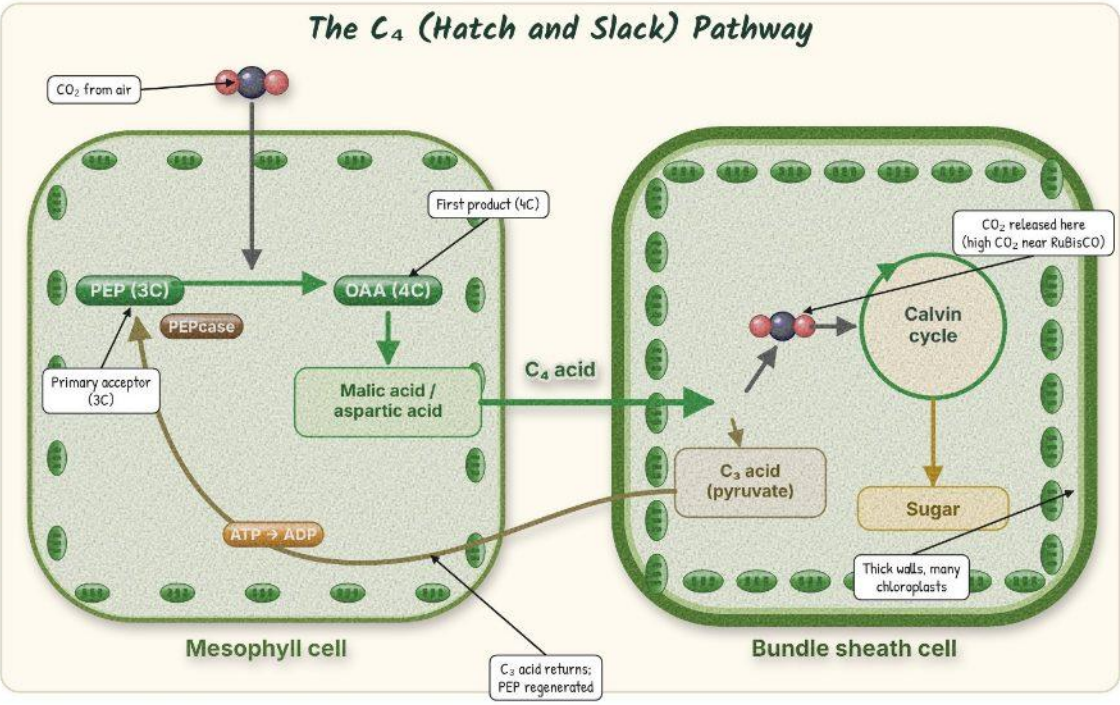
- H^+ high in lumen: water splitting · H carrier · NADP reductase takes H^+ from stroma.
- ★ H^+ return through CF_0; CF_1 changes shape → ATP. ~3 H^+ = 1 ATP.

6. Calvin Cycle



- ★ Calvin, ^{14}C · acceptor RuBP (5C) · enzyme RuBisCO · 1st product 3-PGA.
- Carboxylation → Reduction → Regeneration.
- ★ Per CO_2: 3 ATP + 2 NADPH; glucose: 18 ATP + 12 NADPH, 6 turns.

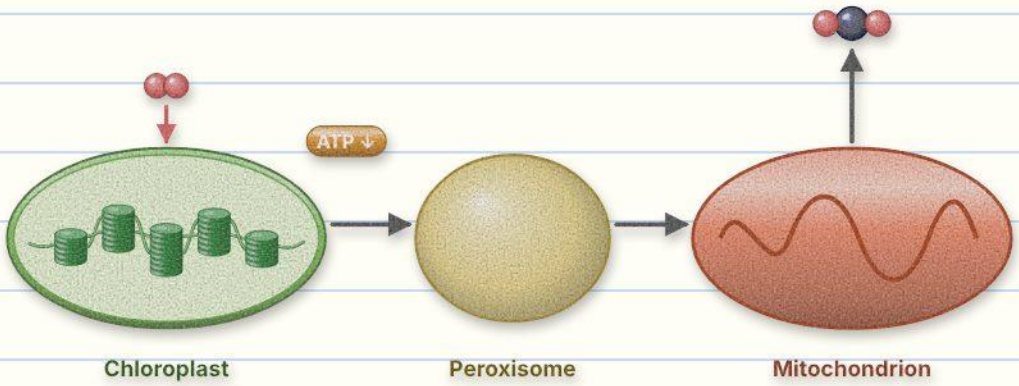
7. C_4 Pathway



- ★ Hatch & Slack · maize, sugarcane · Kranz anatomy (wreath of bundle sheath cells).
- Mesophyll: PEP + CO_2 → OAA (PEPcase; no RuBisCO).
- Bundle sheath: C_4 acid → CO_2 → Calvin cycle; C_3 acid returns → PEP.

8. Photorespiration

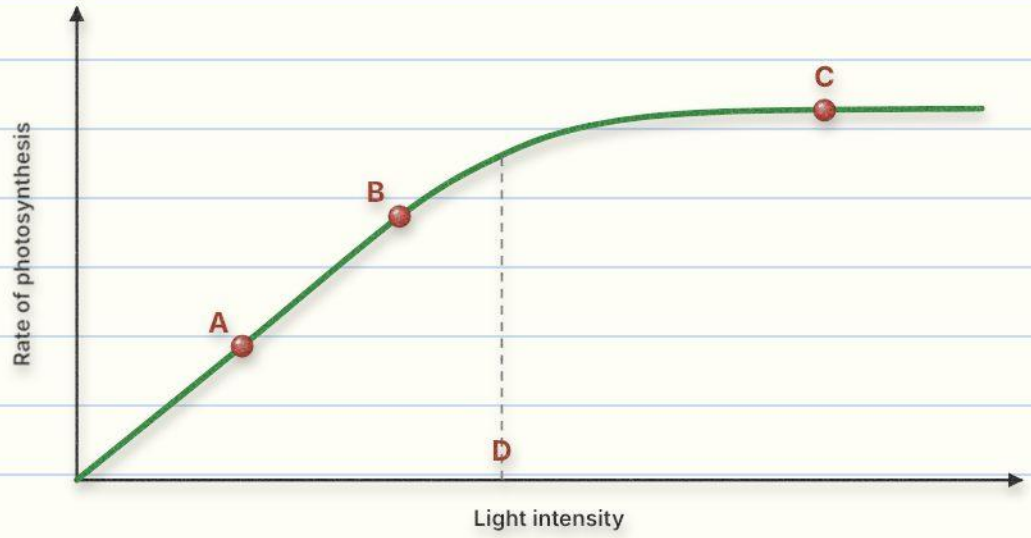
- ★ RuBisCO binds O_2: RuBP → 1 PGA + 1 phosphoglycolate (2C) → C_2 pathway.
- No sugar, no ATP; ATP used, CO_2 released → wasteful.
- ★ Absent in C_4 — high CO_2 near RuBisCO.



	C_3	C_4
Acceptor	RuBP	PEP
1st product	3-PGA	OAA
Photorespiration	yes	no
Opt. temp	10–25 °C	30–45 °C

9. Factors

★ Blackman (1905): rate is set by the factor nearest its minimum.



- Light: saturation at ~10% sunlight · too much → solarisation.
- ★ CO_2 (0.03–0.04%) = major limiting factor; C_3 crops (tomato, capsicum) gain in CO_2-rich greenhouses.
- Temperature controls dark reactions · water stress closes stomata.